

Exterior & Siding / Cladding Products

Introduction & Purpose

Builders FirstSource (BFS) supplies a wide range of siding and cladding systems—from traditional vinyl and cedar to modern fiber-cement, steel and composite materials. Siding/cladding protects the structure from weather and pests, adds curb appeal and can influence energy efficiency. This knowledge article introduces common siding/cladding materials, highlights key features such as thickness, fire resistance and warranties, and summarizes typical installation practices. It is designed to help BFS associates and customers choose the right exterior material for a particular climate or project, understand trade-offs, and identify opportunities for innovation.

Types of Siding / Cladding

Vinyl siding

Vinyl siding is made from poly-vinyl chloride (PVC). Standard vinyl siding thickness ranges from **0.035 in** (minimum standard) to **0.048 in** for premium panels ¹. Regular vinyl siding has an **R-value ≈ 0.61** and **insulated vinyl siding** (vinyl panels with rigid foam backing) reaches **R-values of 2.0–4.0** ². Vinyl products resist moisture and pests and never require painting. However, they can melt or deform at temperatures above $\sim 730^\circ\text{F}$ (389°C) ³ and therefore generally achieve only **Class B or C fire ratings** ⁴.

Dimensions & profiles. Horizontal panels are typically **12 ft** long and offered in widths from **4 in to 8 in**, giving exposures of 3–7 in. Manufacturers like Alside list thickness by product: Board-and-batten vinyl siding (0.048 in), Charter Oak (0.046 in), Conquest (0.040 in) and others ¹.

Installation considerations. Vinyl expands and contracts significantly with temperature; installers must leave a **$\frac{1}{4}$ in gap** between siding and corner posts/channels (increase to $\frac{3}{8}$ in in cold weather) and never fasten panels tight—nail heads should remain $\frac{1}{32}$ in away so panels can move ⁵. Panels overlap 1 in–1 $\frac{1}{4}$ in and should not be stretched during installation ⁶. Around windows and doors, maintain a **$\frac{1}{4}$ in gap** and use trim channels to conceal edges ⁷.

Pros & cons (CoT).

- **Pros:** Low cost, wide color/texture selection, lightweight, minimal maintenance, higher R-value when insulated ².
- **Cons:** Lower fire resistance (Class B/C) and can melt under high heat ³; thermal expansion necessitates careful installation ⁵; may crack in impact/hail; limited longevity vs. composites.

Fiber-cement siding

Fiber-cement siding (e.g., HardiePlank) is composed of cement, cellulose fibers and sand. It is sold in **12-ft (144 in) planks** with nominal widths from **5 $\frac{1}{4}$ in to 12 in**; exposures range from **4 in to 10 $\frac{3}{4}$ in** ⁸.

Thickness is typically **5/16 in (0.312 in)** ⁹. Fiber-cement siding has an **R-value ≈ 0.37** ¹⁰ and is non-combustible, giving **Class A fire ratings** and resisting flame spread. James Hardie products carry a **30-year non-prorated substrate warranty** with a **15-year finish warranty** ¹¹.

Installation considerations. HardiePlank can be installed over wood or steel studs or OSB sheathing with a maximum spacing of 24 in on center ¹². A water-resistive barrier is required ¹³, and planks should not contact standing water or be used as fascia ¹⁴. Install a **1¼ in starter strip** to set the angle and space fasteners at 24 in OC ¹⁵. Leave a small gap between plank ends and trim and fill with flexible caulk ¹⁶; butt joints should be in moderate contact ¹⁶. Store planks flat and keep dry; cutting requires dust-control equipment ¹⁷.

Pros & cons (CoT).

- **Pros:** Class A fire rating, very durable, resists rot and pests, replicates wood appearance, long warranty ¹¹.
- **Cons:** Heavier and more brittle than vinyl, requires special cutting tools and respirator; lower insulating value; needs painting or factory-colored product; installation cost is higher.

Wood siding

Traditional wood siding such as **Western Red Cedar bevel** is produced by re-sawing lumber at an angle so each board has a thick butt and thin top ¹⁸. Common nominal dimensions include **½ × 4 in**, **½ × 6 in**, **½ × 8 in**, **1¼ × 6 or 8 in**, **¾ × 6–10 in**, and larger proprietary sizes ¹⁹. Wood bevel siding creates deep shadow lines and offers an R-value of $\approx 0.8 - 1.2$ per inch of thickness (not cited) but requires regular maintenance (staining/painting). Cedar is naturally resistant to rot and insects; however, wood is combustible (Class C fire rating), may swell with moisture and is susceptible to warping or splitting.

Pros & cons (CoT).

- **Pros:** Natural appearance, renewable resource, easier to cut and install than fiber cement; certain species (cedar, cypress) have natural rot resistance.
- **Cons:** Requires regular finishing; combustible; can warp, split or cup; limited board lengths (usually 16 ft or less) increase waste; lower dimensional stability in humid climates.

Steel/metal siding

Steel siding panels are roll-formed from galvanized or Galvalume® steel. Panel thickness is expressed in gauge; **24 gauge** and **26 gauge** are standard, with **24 gauge recommended for harsh climates** (heavy snow, hail or high winds) ²⁰ ²¹. Thicker gauges resist denting and oil canning but cost more. Panel profiles include board-and-batten, corrugated, flush and lap styles. Metal siding is non-combustible and often rated Class A for fire resistance. Premium finishes like PVDF (Kynar 500®) resist fade and chalk. Manufacturers such as **EDCO** provide a **lifetime, non-prorated warranty** on steel siding with **50-year hail protection**, and guarantee the color finish against fading for **at least 35 years** ²². The ENTEx® finish uses ceramic pigments and reflective technology to reduce thermal gain ²³.

Pros & cons (CoT).

- **Pros:** Fire-resistant (Class A), very low maintenance, durable against hail and wind, long warranties (lifetime material; color fade warranties up to 35 yrs) ²² .
- **Cons:** Higher initial cost; gauge selection affects cost and performance; may require furring strips to prevent galvanic corrosion when installed over treated lumber; denting possible; thermal conductivity can transmit noise or heat unless insulated.

Composite / engineered siding

Composite siding encompasses engineered wood, PVC and new hybrid systems.

Fiber-cement composite (Aspyre/Hardie)

Already covered under fiber-cement section.

Poly-ash (TruExterior)

TruExterior® siding uses a poly-ash formulation (70 % recycled polymers and fly-ash). Boards are formed in continuous processes and cut to **16-ft lengths**. The nominal **thickness is ¾ in**, with standard **widths of 5½ in, 7½ in, 9½ in and 11½ in** ²⁴ . Poly-ash siding has a coefficient of linear expansion **<1.40 × 10⁻⁵ in/in/°F** and water absorption **< 1.5 %** ²⁵ , making it dimensionally stable. TruExterior is backed by a **20-year warranty** ²⁶ and is rated for ground and masonry contact ²⁷ . Boards require field painting within 150 days of installation ²⁸ .

Pros & cons (CoT).

- **Pros:** High dimensional stability (low expansion and water absorption), termite resistant, rated for ground contact, 20-year warranty ²⁶ .
- **Cons:** Heavier than vinyl, must be painted, non-structural; cost is higher than fiber cement; requires skilled installation.

PVC / polymer cladding

PVC cladding (e.g., AZEK/Westlake Royal) is extruded cellular PVC or polymer boards. Products include shingle-style panels, board-and-batten planks and trim. PVC siding is lighter than fiber-cement and has high moisture resistance. It can meet Class A fire ratings when formulated with fire retardants; however, many products carry Class B. Warranty periods range from 25 – 50 years.

Composite foam-polymer (ASCEND)

Alside's **ASCEND® composite cladding** consists of glass-reinforced polymer and graphite-infused polystyrene. Each plank is **¾ in thick, 8.25 in wide with 7 in exposure**, length **12 ft 3 in**, butt height **¾ in**, and weight **0.31 lb/ft** ²⁹ . It has an **R-value of 2.0**, wind-load resistance of **53 psf** (≈180 mph), impact resistance of **57 in-lb** and coefficient of linear expansion **2.3 × 10⁻⁵ in/in/°F** ³⁰ . Alside provides a **lifetime limited warranty** against blistering, corroding, flaking and peeling ³¹ , with **lifetime fade protection** and transferability provisions ³² .

Pros & cons (CoT).

- **Pros:** High R-value (≈ 2), Class A fire rating, lightweight, long warranty, engineered to resist fade, hail and high winds ²⁹ ³¹ .
- **Cons:** Newer technology with higher cost; limited color palette; must be installed per manufacturer guidelines to maintain warranty.

Tree-of-Thought reasoning by climate zone

When choosing siding, branch first by **climate** (cold/hot, humid/dry, wildfire risk) and then by **desired attributes** (maintenance, cost, aesthetics):

1. **Coastal/High humidity:** Resist moisture, salt and wind. Fiber-cement, steel and poly-ash composites perform well; vinyl may warp under extreme heat and wind; cedar requires frequent maintenance.
2. **Cold climates with freeze-thaw cycles:** Materials should resist moisture intrusion and expansion. Vinyl can crack in extreme cold; fiber-cement and poly-ash maintain stability; composite systems with insulating foam (ASCEND) improve thermal comfort.
3. **Hot/dry or wildfire-prone regions:** Non-combustible materials (fiber-cement, steel) or Class A composite products are preferred to reduce fire risk ⁴ . PVC and vinyl may melt; wood is combustible.
4. **High-wind/hail zones:** Thick-gauge steel or reinforced composite panels provide superior impact resistance; fiber-cement resists moderate wind but can crack if impacted; vinyl may blow off if not properly installed.

Trim, Flashing, Soffit & Fascia Components

Proper cladding installation includes several accessories and flashings to manage water and create a finished look.

- **Trim boards:** Surround windows, doors and corners; provide crisp edges and hide cut panel ends. Fiber-cement, poly-ash and PVC trim are available. For example, **Plycem fiber-cement trim** resists rot and insects ³³ .
- **Flashing:** Thin material (aluminum, vinyl or membrane) placed behind J-channels, corner posts and over window heads to prevent water infiltration ³⁴ . Drip caps/head flashing protect vertical siding or panel joints ³⁵ .
- **Soffit:** Panels used to enclose the underside of eaves. They may be vented for attic ventilation. Materials include vinyl, aluminum, fiber-cement or PVC ³⁵ .
- **Fascia:** Trim covering rafter ends at the roof edge ³⁵ . Fascia boards can be wrapped with aluminum coil or PVC for low maintenance; fascia “caps” snap over wood fascia ³⁴ .
- **Starter strip:** Metal or vinyl strip installed at the base of the wall to lock the first siding course ⁶ ¹⁶ .
- **J-channel/utility trim:** Receives siding edges around windows, doors and soffit; provides expansion space ⁷ .

Specification Factors (weather, fire, moisture, expansion)

Factor	Vinyl	Fiber-cement	Wood	Steel/Metal	Composite (poly-ash, ASCEND)
Fire rating	Class B/C; melts at ≈730 °F ³	Class A; non-combustible	Class C; combustible	Class A; non-combustible	Poly-ash: non-combustible (Class A); ASCEND: Class A fire rating ³⁰
Moisture resistance	Excellent if seams are flushed; may trap moisture behind panels	Resists rot & termites; requires gap & caulk ¹⁶	Good (cedar) but susceptible to swelling/ decay without maintenance	Excellent; galvanized coating resists rust; finish protects against corrosion ²²	Poly-ash absorbs <1.5 % water ²⁵ ; ASCEND is moisture-resistant and has 2.0 R-value ³⁰
Thermal expansion	High; leave ¼ in gap; allow panel movement ⁵	Moderate; butt joints in light contact; gap & caulk at trim ¹⁶	Moderate to high, dependent on grain orientation	Low; metal expands but less than vinyl; gauge affects movement	Poly-ash coefficient <1.40×10 ⁻⁵ in/in/ °F ²⁵ ; ASCEND coefficient 2.3×10 ⁻⁵ in/in/°F ³⁰
Insulating value (R-value)	≈0.61 (regular); 2.0–4.0 (insulated) ²	≈0.37 ¹⁰	≈0.8–1.2 per inch	Low unless insulated backing	2.0 (ASCEND) ³⁰ ; poly-ash similar to wood
Warranty	Varies; 20– 40 yrs; lifetime for premium vinyl (prorated)	30-yr substrate & 15-yr finish ¹¹	10–15 yrs (varies by species/ finish)	Lifetime, non-prorated with 50-yr hail protection ²²	Poly-ash: 20-yr ²⁶ ; ASCEND: lifetime limited ³¹

Note: R-values for wood and steel are approximate; actual insulation value depends on wall assembly.

Typical Dimensions & Material Trade-offs

Material	Typical length	Width/exposure	Thickness	Weight	Trade-offs
Vinyl siding	12 ft panels	4–8 in widths; exposures 3–7 in	0.035–0.048 in ¹	~1 lb/ft ²	Low cost, wide color choices; lower fire rating; high thermal expansion; moderate impact resistance
Fiber-cement lap (HardiePlank)	12 ft (144 in) planks	5¼–12 in widths; exposures 4–10¾ in ⁸	5/16 in ⁹	~2.3 lb/ft ²	Fire-resistant, durable; heavy; requires special tools; moderate insulation
Western Red Cedar bevel	Up to 16 ft	Nominal sizes ½ × 4–8 in, 1⅞ × 6–8 in, ¾ × 6–10 in ¹⁹	Varies: ½–¾ in ¹⁹	1.3 lb/ft ² (approx.)	Natural beauty; sustainable; combustible; requires maintenance; limited lengths
Steel siding	12–16 ft panels	Board-and-batten widths vary by profile; exposures 6–8 in	Gauges 24–26 (thicker gauge = lower number) ^{20 21}	~1–1.5 lb/ft ²	Non-combustible; long life; high impact resistance; cost depends on gauge and finish
Poly-ash (TruExterior)	16 ft boards	Widths 5½, 7½, 9½, 11½ in ²⁴	¾ in ²⁴	~2.2 lb/ft ²	Stable, termite-resistant; 20-yr warranty ²⁶ ; must be painted; non-structural
ASCEND composite	12 ft 3 in	8.25 in width; 7 in exposure ²⁹	¾ in (butt ⅝ in) ²⁹	0.31 lb/ft (≈0.5 lb/ft ²) ²⁹	High R-value (2.0), Class A fire rating, lifetime warranty; requires proper installation to maintain warranty ³¹

Installation & Best Practices

1. **Start with a solid substrate.** All siding should be installed over plumb and properly sheathed walls. Fiber-cement and composite products require a **water-resistive barrier** behind the siding ¹³ ; vinyl siding relies on the housewrap to shed water.
2. **Provide expansion gaps.**
3. Vinyl: leave a **¼ in gap** at joints and accessories (³/₈ in in cold weather) and never nail tight; fasteners should allow ¹/₃₂ in movement ⁵ .
4. Fiber-cement: butt joints in **moderate contact**; leave a small gap between planks and trim then caulk ¹⁶ .
5. Poly-ash and composite systems: follow manufacturer-specified gaps; ASCEND installation guidelines emphasize accommodating thermal expansion to preserve warranty ³¹ .
6. **Use starter strips and flashings.** Install a **starter strip** at the base of the wall to support the first course ⁶ ¹⁶ . Use **drip cap/head flashing** above doors and windows and behind vertical trim to direct water away ³⁵ .
7. **Fasten correctly.**
8. Vinyl: Use corrosion-resistant nails or screws; drive fasteners through the center of nailing slots; do not overdrive.
9. Fiber-cement: Nails should penetrate studs or sheathing; place nails ³/₈ in from plank edges ³⁶ and 1 in from the top of the plank if no nail line is present ³⁷ .
10. Steel: Use manufacturer-specified screws with rubber washers; avoid galvanic contact with treated wood.
11. **Cut safely and manage dust.** Cutting fiber-cement produces crystalline silica; use HardieBlade® saw blades with dust collection or shears ¹⁷ . Vinyl and poly-ash can be cut with standard carbide blades; metal requires metal-cutting saws or snips.
12. **Seal and paint as required.** Poly-ash boards must be painted within 150 days of installation ²⁸ . Fiber-cement planks with ColorPlus® finish should not be field-painted except at cut edges; Hardie offers a 15-yr finish warranty ¹¹ .

SCAMPER Innovation Ideas

- **Substitute:** Replace traditional vinyl with **composite cladding** that integrates insulated foam to improve R-value and reduce thermal bridging.
- **Combine:** Pair **solar cladding panels** with steel or composite siding to generate renewable energy; integrate rain-screen battens to allow ventilation behind siding.
- **Adapt:** Use **poly-ash trim** as a moisture-resistant substitute for wood in ground-contact applications; adapt board-and-batten patterns to modern farmhouses using composite boards ³⁸ .
- **Modify:** Incorporate **integrated LED lighting** into aluminum soffit and fascia systems for decorative eaves lighting.
- **Put to other use:** Use **steel panels** with high solar reflectance to reduce cooling loads; salvage cedar siding for interior accent walls.
- **Eliminate:** Design cladding systems with **fewer seams** by using larger composite panels or structural insulated panels; this reduces potential water intrusion points.
- **Rearrange/Re-engineer:** Develop **modular cladding cartridges** where insulation, weather barrier and finish panel are assembled off-site for rapid installation.

Likely User Questions

- **Is fiber-cement better than vinyl in coastal regions?** Fiber-cement resists salt air, high winds and offers a Class A fire rating, making it a strong option for coastal areas; however, it weighs more and costs more than vinyl.
- **What expansion gap is required for vinyl siding?** Leave $\frac{1}{4}$ in at each end ($\frac{3}{8}$ in when temperatures are below 40 °F) ⁵ ; do not nail panels tight.
- **How do you flash around windows in siding?** Install head flashing/drip cap above windows, wrap rough openings with water-resistive barrier, and insert J-channels to receive cut siding edges ³⁵ .
- **Does steel siding get hot?** Coatings such as EDCO's ENTEx incorporate reflective pigments to minimize heat gain ²³ ; steel often performs better than vinyl in heat resistance.
- **What is the warranty for ASCEND composite cladding?** Alside provides a **lifetime limited warranty** covering manufacturing defects and lifetime fade protection ³¹ ³² .
- **Can poly-ash siding touch the ground?** Yes; TruExterior boards are rated for ground and masonry contact and carry a 20-year warranty ³⁹ .
- **How thick is fiber-cement trim?** Fiber-cement trim, such as Plycem, typically ranges from $\frac{3}{4}$ in to 1 in thick and is resistant to rot and warping ³³ .
- **Does vinyl siding provide insulation?** Regular vinyl offers an R-value of ~0.61, while insulated vinyl (with rigid foam backing) has R-values between **2 and 4** ² .

Related Topics / Crosslinks

- **Roofing materials** (asphalt shingles, metal roofs) and how roofing and siding interact with flashing and vented soffits.
- **Weatherization & flashing** (housewrap, window flashing tapes). See BFS category: Weatherization & Flashing.
- **Windows & doors** (trim integration, head flashing) and **moulding & millwork** for complementary products.
- **Decking & railing** (transition details where siding meets decks and patios).
- **Insulation & energy efficiency** (use of continuous insulation or insulated siding to improve the building envelope).

Note: The information above summarizes publicly available manufacturer data, installation manuals and industry articles. Always consult the specific manufacturer's installation instructions and local building codes when selecting and installing siding or cladding products.

¹ Vinyl Siding Thickness: Does it Matter or Not? - Alside
<https://www.alside.com/blog/vinyl-siding-thickness-does-it-matter-or-not/>

² ¹⁰ Understanding the R-Value Of Siding | Garden State Brickface and Siding
<https://www.brickface.com/understanding-the-r-value-of-siding/>

³ ⁴ Is Vinyl Siding Fire-Resistant? | DaBella
<https://dabella.us/2025/06/is-vinyl-siding-fire-resistant/>

5 6 7 **Variform-Siding-Installation-Guide.pdf**

<https://www.plygem.com/wp-content/uploads/shared/Widen%20Assets/Variform%20Website/Installation%20Guides/Variform-Siding-Installation-Guide.pdf>

8 **HardiePlank Lap Siding - Product Data**

<https://www.buildsite.com/pdf/jameshardie/HardiePlank-Lap-Siding-Product-Data-1987200.pdf>

9 **Hardie® Panel 215607 Fiber Cement Lap Siding,**

<https://mccoys-website.useast01.umbraco.io/shop/p/02129931/hardiepanel-215607-fiber-cement-lap-siding-prime-coated-12-in-w-144-in-l-0312-in-thickness>

11 **Lifetime Value Siding Products | James Hardie**

<https://www.jameshardie.com/why-james-hardie/lifetime-value-siding-products/>

12 13 14 15 16 17 36 37 **HardiePlank Lap Siding - Installation Instructions**

<https://www.buildsite.com/pdf/jameshardie/HardiePlank-Lap-Siding-Installation-Instructions-1987199.pdf>

18 19 **Western Red Cedar Bevel Siding - Real Cedar**

<https://www.realcedar.com/siding/profiles/bevel-siding/>

20 21 **What Gauge Should I Use For Metal Siding?**

<https://www.westernstatesmetalroofing.com/blog/metal-siding-gauge>

22 23 **EDCO Steel Siding Offers Quality Home Protection**

https://www.edcoproducts.com/articles/Steel_Siding_Quality_Protection.html

24 25 28 **Boral TruExterior Siding**

https://www.fp-supply.com/cmss_files/attachmentlibrary/truexterior_3part_spec_siding.pdf

26 27 38 39 **TruExterior Siding 101: Everything You Need to Know - U.S. LUMBER**

<https://www.uslumber.com/truexterior-siding-101-everything-you-need-to-know/>

29 30 **ahrb-3-28-2022-25-maple-street-ascend-technical-summary**

<https://www.dobbsferry.gov/building-department/files/ahrb-3-28-2022-25-maple-street-ascend-technical-summary>

31 32 **ASCEND-Composite-Cladding-Warranty**

<https://www.alside.com/wp-content/uploads/sites/3/2024/12/ASCEND-Composite-Cladding-Warranty.pdf>

33 **Siding & Exterior Trim - Builders FirstSource - NE eShowroom**

<https://bfs-ne.myeshowroom.com/category/siding-trim>

34 35 **Vinyl Siding Terminology with Pictures | PEPA**

<https://polymericexteriors.org/installation/installation-manual/terms-to-know/>